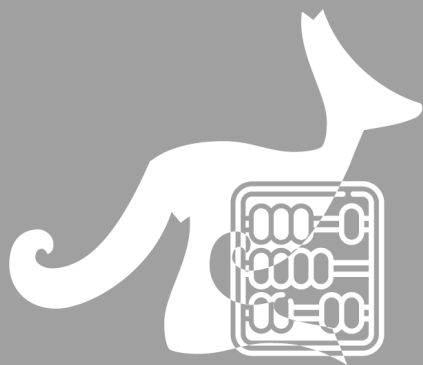




ASDAN | 阿思丹

China

学院



MATH KANGAROO CHINA

考试说明 / Instruction

Time allowed: 75 minutes

考试时长: 75 分钟

Language: English & Chinese

试卷语言: 英文和中文

This is a 30-question multiple-choice competition.

考试包含 30 道单项选择题。

Problems 1 to 10 are worth 3 points each.

1-10 题每题 3 分

Problems 11 to 20 are worth 4 points each.

11-20 题每题 4 分

Problems 21 to 30 are worth 5 points each.

21-30 题每题 5 分

1 point will be deducted for an incorrect answer and no point for a blank answer.

答错扣 1 分, 不答不扣分。

The full score is 150 (Starting point is 30)

总分为 150 分 (起始分数为 30 分)

Do not open this booklet until you are told to do so.

收到监考老师指令后, 方可打开试卷册。

Mark your answers clearly on the answer sheet using a

2B pencil.

务必使用 **2B 铅笔** 在答题卡上正确填涂自己的答案。

Calculator cannot be used during the competition.

考试期间禁止使用计算器。

Good Luck!

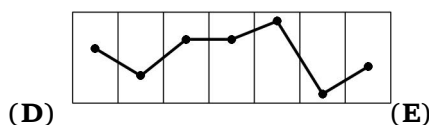
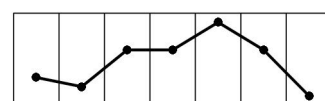
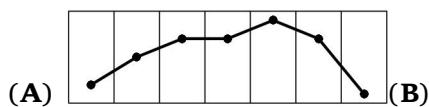
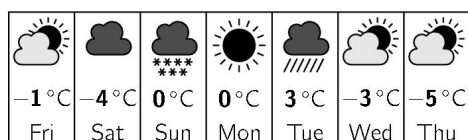
预祝考试顺利

**Year 11
&
Year 12**

3 points

1. Paula's weather app shows a diagram of the predicted weather and maximum temperatures for the next seven days, as shown. Which of the following represents the corresponding graph of maximum temperatures?

Paula 的天气应用软件上显示了未来七天的天气预报和每日最高温度，如下图。请问下列哪副图可以表示未来七天的每日最高温度走势？



2. How many integers are in the interval $(20 - \sqrt{21}, 20 + \sqrt{21})$?

区间 $(20 - \sqrt{21}, 20 + \sqrt{21})$ 中有多少个整数？

- (A) 9 (B) 10 (C) 11 (D) 12 (E) 13

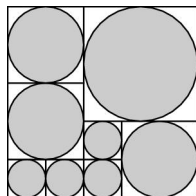
3. A cube with edge 1 is cut into two identical cuboids. What is the surface area of one of these cuboids?

将一个边长为 1 的立方体切割成两个完全相同的长方体。每个长方体的表面积是多少？

- (A) $\frac{3}{2}$ (B) 2 (C) 3 (D) 4 (E) 5

4. A large square is divided into smaller squares, as shown. A shaded circle is inscribed inside each of the smaller squares. What proportion of the area of the large square is shaded?

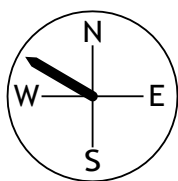
将一个正方形分成若干个小正方形，如下图。每个小正方形内都有一个被涂成阴影的内切圆。请问大正方形中阴影部分的面积占比是多少？



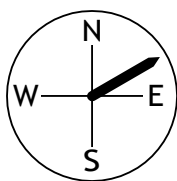
- (A) $\frac{8\pi}{9}$ (B) $\frac{13\pi}{16}$ (C) $\frac{3}{\pi}$ (D) $\frac{3}{4}$ (E) $\frac{\pi}{4}$

5. After the storm last night, the flagpole on our school building is leaning over. Looking from northwest, its tip is to the right of its bottom point. Looking from the east, its tip is also to the right of its bottom point. In which direction could the flagpole be leaning over?

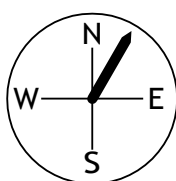
受昨天夜里暴风雨的影响，学校的旗杆发生了倾斜。从西北方向看，旗杆顶端在底部的右侧；从正东方向看，旗杆顶端也在底部的右侧。请问旗杆向哪个方向倾斜？



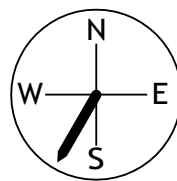
(A)



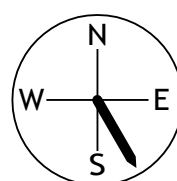
(B)



(C)



(D)



(E)

6. A rectangular sheet of paper has length x and width y , where $x > y$. The rectangle may be folded to form the curved surface of a circular cylinder in two different ways. What is the ratio of the volume of the longer cylinder to the volume of the shorter cylinder?

一张长方形纸张长为 x ，宽为 y ，且 $x > y$ 。有两种不同的方式可以将这张纸围成圆柱体的侧面。

请问围成的较高圆柱体与较矮圆柱体的体积之比是多少？

- (A) $y^2 : x^2$ (B) $y : x$ (C) $1 : 1$ (D) $x : y$ (E) $x^2 : y^2$

7. Let $x = \frac{\pi}{4}$. Which of the following numbers is the largest?

假设 $x = \frac{\pi}{4}$ ，请问下列选项中哪个数最大？

- (A) x^4 (B) x^2 (C) x (D) $\sqrt{x}$ (E) $\sqrt[4]{x}$

8. How many 3-digit-numbers formed using only the digits 1, 3 and 5 are divisible by 3? You may use digits more than once.

只使用数字 1、3、5 组成的三位数中，有多少个可以被 3 整除？每个数字可以重复使用。

- (A) 3 (B) 6 (C) 9 (D) 18 (E) 27

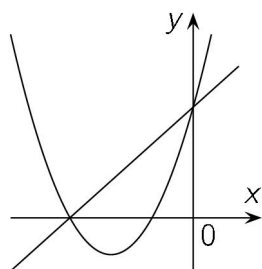
9. What is the area of the triangle whose vertices are at (p, q) , $(3p, q)$ and $(2p, 3q)$, where $p, q > 0$?

三角形三个顶点的坐标分别为 (p, q) , $(3p, q)$ 和 $(2p, 3q)$ ，且 $p, q > 0$ 。请问该三角形的面积是多少？

- (A) $\frac{pq}{2}$ (B) pq (C) $2pq$ (D) $3pq$ (E) $4pq$

10. The parabola in the figure has an equation of the form $y = ax^2 + bx + c$ for some distinct real numbers a, b and c . Which of the following equations could be an equation of the line in the figure?

图中抛物线的方程为 $y = ax^2 + bx + c$ ，其中 a, b 和 c 为不同的实数。请问下列哪个选项中的方程可能是图中直线对应的方程？



- (A) $y = bx + c$ (B) $y = cx + b$ (C) $y = ax + b$ (D) $y = ax + c$ (E) $y = cx + a$

4 points

11. What proportion of all the divisors of $7!$ is odd?

$7!$ 的所有因数中，奇因数占的比例是多少？

- (A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{1}{4}$ (D) $\frac{1}{5}$ (E) $\frac{1}{6}$

12. If $A = (0, 1) \cup (2, 3)$ and $B = (1, 2) \cup (3, 4)$, what is the set of all numbers of the form $a + b$ with a in A and b in B ?

如果集合 $A = (0, 1) \cup (2, 3)$, 集合 $B = (1, 2) \cup (3, 4)$, 已知 a 属于集合 A , b 属于集合 B , 请问下列哪一项是所有 $a + b$ 的值构成的集合?

- (A) $(1, 7)$ (B) $(1, 5) \cup (5, 7)$ (C) $(1, 3) \cup (3, 7)$
(D) $(1, 3) \cup (3, 5) \cup (5, 7)$ (E) none of the previous 以上均不是

13. How many three-digit natural numbers have the property that when their digits are written in reverse order, the result is a three-digit number which is 99 more than the original number?

将一个三位自然数各个数位上的数字倒序写出, 得到的新三位数比原来的三位数大 99。请问这样的三位数共有多少个?

- (A) 8 (B) 64 (C) 72 (D) 80 (E) 81

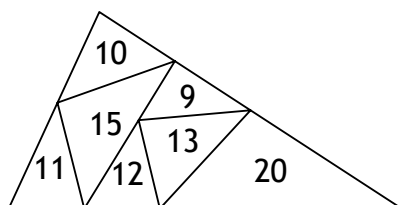
14. The first 1000 positive integers are written in a row in some order and all sums of any three adjacent numbers are calculated. What is the greatest number of odd sums that can be obtained?

将前 1000 个正整数按某种顺序写成一排, 并计算任意三个相邻数的和。请问最多能得到多少个和为奇数?

- (A) 997 (B) 996 (C) 995 (D) 994 (E) 993

15. A large triangle is divided into smaller triangles as shown. The number inside each small triangle indicates its perimeter. What is the perimeter of the large triangle?

一个大三角形被分成若干个小三角形, 如下图。每个小三角形内的数字表示其周长, 请问这个大三角形的周长是多少?



- (A) 31 (B) 34 (C) 41 (D) 62

(E) none of the previous 以上均不正确

16. For a positive integer N , we denote by $p(N)$ the product of the digits of N when written in decimal form. For example, $p(23) = 2 \times 3 = 6$. What is the value of the sum $p(10) + p(11) + p(12) + \dots + p(99) + p(100)$?

对于正整数 N , $p(N)$ 表示 N 的各个数位上数字的乘积, 比如 $p(23) = 2 \times 3 = 6$ 。请问 $p(10) + p(11) + p(12) + \dots + p(99) + p(100)$ 的和是多少?

(A) 2025 (B) 4500 (C) 5005 (D) 5050

(E) none of the previous 以上均不正确

17. In the 5×5 square shown the sum of the numbers in each row and in each column is the same. There is a number in every cell, but some of the numbers are not shown. What is the number in the cell marked with a question mark?

如图所示, 在 5×5 的网格中, 每一行和每一列的数字之和均相同。每个单元格里都有一个数字, 但有些数字没有显示。请问标有问号的单元格的数字是什么?

	16		22	
20		21		2
	25		1	
24		5		6
	4		?	

(A) 8 (B) 10 (C) 12 (D) 18 (E) 23

18. A piece of string is lying on the table. It is partially covered by three coins as seen in the

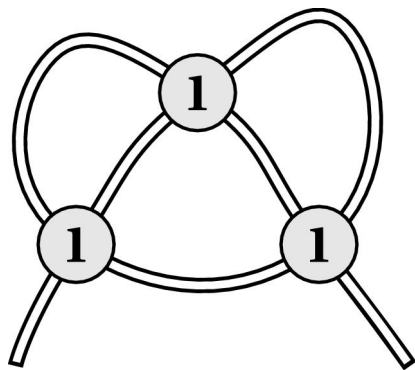


figure.

Under each coin the string is equally likely to pass over itself like this:

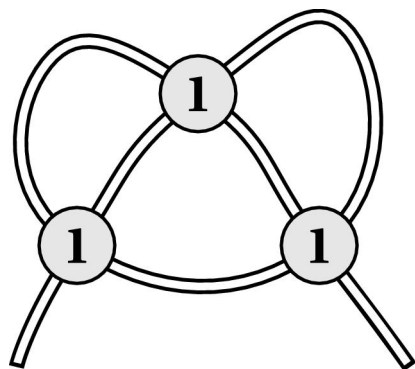


or like this:



What is the probability that the string is knotted after its ends are pulled?

如图所示，一根绳子放在桌子上，三枚硬币盖住了绳子交叉的部分。



在每一枚硬币下方，绳子的交叉方式可能是



或者



拉扯绳子的两端，请问绳子会打结的概率是多少？

(A) $\frac{1}{2}$

(B) $\frac{1}{4}$

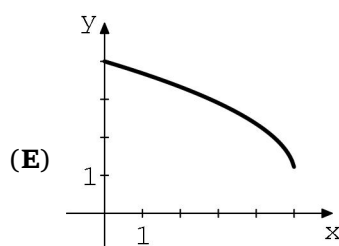
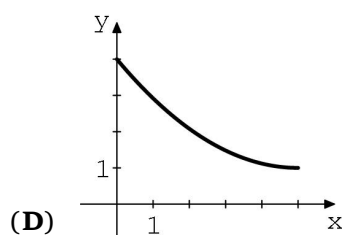
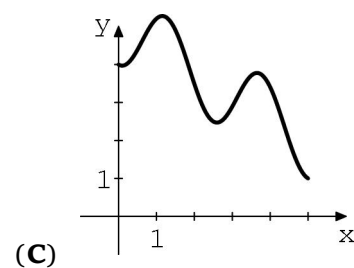
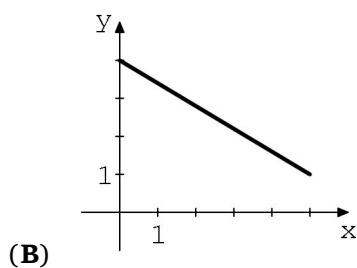
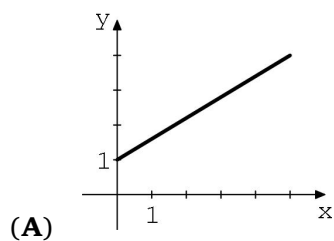
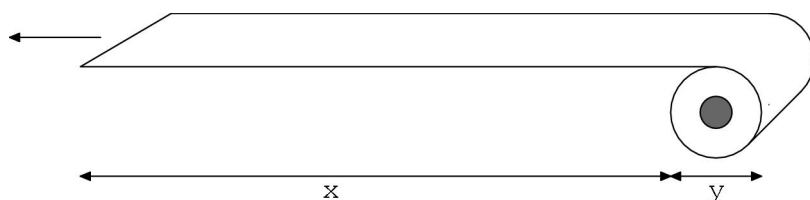
(C) $\frac{1}{8}$

(D) $\frac{3}{4}$

(E) $\frac{3}{8}$

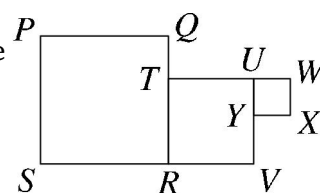
19. A naughty pup grabs the end of a roll of toilet paper and walks away at a constant speed. Which of the functions below best describes the thickness y of the roll as a function of the unrolled part x ?

一只淘气的小狗抓住一卷卫生纸的末端，然后匀速跑走。请问下列哪个函数图像可以表示卫生纸厚度 y 和被铺开长度 x 之间的关系？



20. The diagram shows three squares, $PQRS$, $TRVU$, and $UWXY$.

They are placed together, edge to edge. Points P , T and X lie on the same straight line. The area of $PQRS$ is 36 and the area of $TRVU$ is 16. What is the area of triangle PXV ?



右图有三个正方形 $PQRS$ 、 $TRVU$ 和 $UWXY$ 。它们彼此相邻，边缘重合。点 P 、 T 和 X 在一条直线上。正方形 $PQRS$ 的面积为 36， $TRVU$ 的面积为 16。请问三角形 PXV 的面积是多少？

(A) $14\frac{2}{3}$

(B) $15\frac{1}{3}$

(C) 16

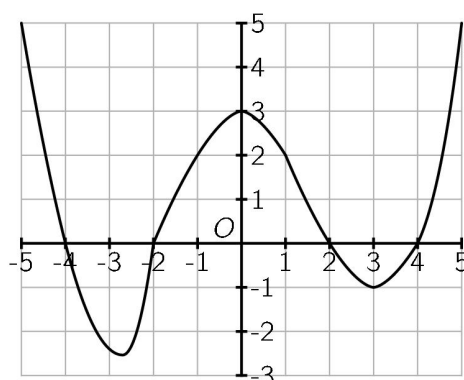
(D) $17\frac{2}{3}$

(E) 18

5 points

21. The figure shows the graph of a function $f: [-5, 5] \rightarrow \mathbb{R}$. How many distinct solutions does the equation $f(f(x)) = 0$ have?

下图为函数 $f: [-5, 5] \rightarrow \mathbb{R}$ 的图像。请问方程 $f(f(x)) = 0$ 有多少个不同的解？



(A) 2

(B) 4

(C) 6

(D) 7

(E) 8

22. The numbers 1, 2, 7, 9, 10, 15 and 19 are written down on a blackboard. Two players alternately delete one number each until only one number remains on the blackboard. The sum of the numbers deleted by one of the players is twice the sum of the numbers deleted by the other player. What is the number that remains?

在黑板上写下数字 1, 2, 7, 9, 10, 15 和 19。两名同学轮流擦除一个数字，直到黑板上只剩下一个数字。其中一名同学擦除的数字之和是另一名同学擦除数字之和的两倍。请问黑板上剩下的数字是多少？

- (A) 7 (B) 9 (C) 10 (D) 15 (E) 19

23. The function $f(x)$ is such that $f(x+y) = f(x) \cdot f(y)$ and $f(1) = 2$. What is the value of

$$\frac{f(2)}{f(1)} + \frac{f(3)}{f(2)} + \dots + \frac{f(2021)}{f(2020)} ?$$

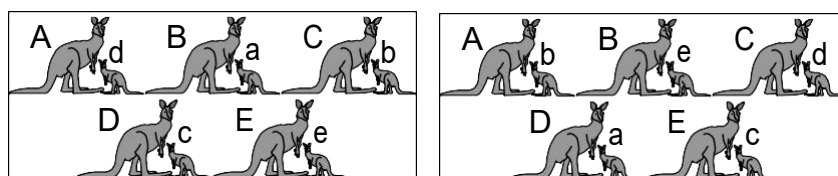
函数 $f(x)$ 满足 $f(x+y) = f(x) \cdot f(y)$ ，且 $f(1) = 2$ 。请问 $\frac{f(2)}{f(1)} + \frac{f(3)}{f(2)} + \dots + \frac{f(2021)}{f(2020)}$ 的值是多少？

- (A) 0 (B) $\frac{1}{2}$ (C) 2 (D) 2020

(E) none of the previous 以上都不正确

24. Five kangaroos named A, B, C, D and E have one child each, named a, b, c, d and e. In the first group photo shown exactly two of the children are standing next to their mothers. In the second group photo exactly three of the children are standing next to their mothers. Whose child is a?

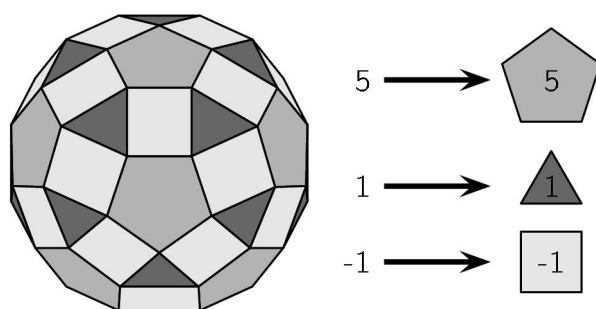
A、B、C、D、E 五只袋鼠各有一个袋鼠宝宝，命名为 a、b、c、d、e。第一组图片中，恰好有两只袋鼠宝宝站在袋鼠妈妈旁边；第二组图片中，恰好有三只袋鼠宝宝站在袋鼠妈妈旁边。请问哪只袋鼠的宝宝是 a？



- (A) A (B) B (C) C (D) D (E) E

25. The solid shown in the diagram has 12 regular pentagonal faces, the other faces being either equilateral triangles or squares. Each pentagonal face is surrounded by 5 square faces and each triangular face is surrounded by 3 square faces. John writes 1 on each triangular face, 5 on each pentagonal face and -1 on each square. What is the total of the numbers written on the solid?

下图所示的立体图形共有 12 个面是正五边形，其他面是等边三角形或正方形。每个五边形面都被 5 个正方形面包围，每个三角形面都被 3 个正方形面包围。John 在每个三角形面上都写上数字 1，在每个五边形面上都写上数字 5，在每个正方形面上都写上数字 -1。请问写在该立体图形上的数字总和是多少？



(A) 20

(B) 50

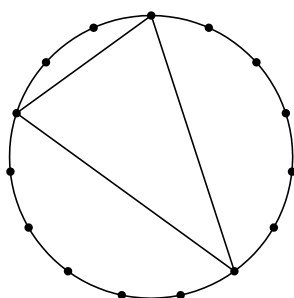
(C) 60

(D) 80

(E) 120

26. On a circle 15 points are equally spaced. We can form triangles by joining any three of these. We count two triangles as being the same if they are congruent i.e. one is a rotation and/or a reflection of the other. How many different triangles can be drawn?

一个圆周上有 15 个等距的点。连接其中任意三个点可以构成三角形。如果两个三角形全等，即其中一个三角形可由另一个旋转变换或反射变换得到，我们就认为这两个三角形是相同的。请问可以画出多少个不同的三角形？



(A) 19

(B) 91

(C) 46

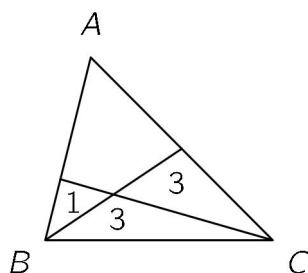
(D) 455

(E) 23

27. A triangle ABC is divided into four parts by two straight lines, as shown. The areas of the smaller triangles are 1, 3 and 3. What is the area of the original triangle?

如图所示，三角形 ABC 被两条直线分为四部分。其中三个小三角形的面积分别为 1、3、3。

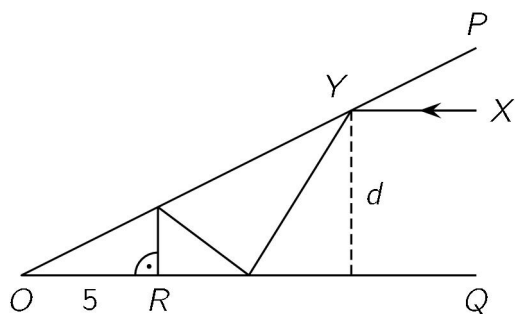
请问大三角形 ABC 的面积是多少？



- (A) 12 (B) 12.5 (C) 13 (D) 13.5 (E) 14

28. Two plane mirrors OP and OQ are inclined at an acute angle (diagram is not to scale). A ray of light XY parallel to OQ strikes mirror OP at Y . The ray is reflected and hits mirror OQ , is reflected again and hits mirror OP and is reflected for a third time and strikes mirror OQ at right angles at R , as shown. The distance OR is 5 cm. The ray XY is d cm from the mirror OQ . What is the value of d ?

如图所示，两个平面镜 OP 和 OQ 倾斜形成一个锐角（图像不是按比例绘制）。平行于镜面 OQ 的光线 XY 射向镜面 OP ，并与 OP 交于点 Y 。光线发生反射后射向 OQ ，再次反射后射向 OP ，最后反射垂直射向 OQ ，并与 OQ 交于点 R 。已知 OR 长 5 厘米，光线 XY 与镜面 OQ 相距 d 厘米。请问 d 的值是多少？



- (A) 4 (B) 4.5 (C) 5 (D) 5.5 (E) 6

29. Let $M(k)$ be the maximum value of $|4x^2 - 4x + k|$ for x in the interval $[-1, 1]$, where k can be any real number. What is the minimum possible value of $M(k)$?

对于 x ，假设 $|4x^2 - 4x + k|$ 在 $[-1, 1]$ 区间内的最大值为 $M(k)$ ，其中 k 为任意实数。请问 $M(k)$ 的最小可能值是多少？

- (A) 4 (B) 9/2 (C) 5 (D) 11/2 (E) 8

30. A certain game is won when one player gets 3 points ahead. Two players A and B are playing the game and at a particular point, A is 1 point ahead. Each player has an equal probability of winning each point. What is the probability that A wins the game?

某场比赛中，选手领先 3 分即可获胜。选手 A 和选手 B 正在进行比赛，在比赛的某个节点，A 领先 1 分。已知每个选手得到每一分的概率都相同。请问选手 A 获胜的概率是多少？

(A) $\frac{1}{2}$

(B) $\frac{2}{3}$

(C) $\frac{3}{4}$

(D) $\frac{4}{5}$

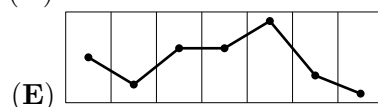
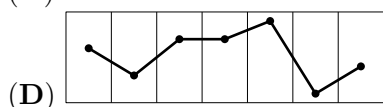
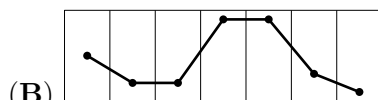
(E) $\frac{5}{6}$

PROBLEMS WITH SOLUTIONS

3 points

1. Paula's weather app shows a diagram of the predicted weather and maximum temperatures for the next seven days, as shown. Which of the following represents the corresponding graph of maximum temperatures?

-1°C	-4°C	0°C	0°C	3°C	-3°C	-5°C
Fri	Sat	Sun	Mon	Tue	Wed	Thu



SOLUTION:

2. How many integers are in the interval $(20 - \sqrt{21}, 20 + \sqrt{21})$?

- (A) 9 (B) 10 (C) 11 (D) 12 (E) 13

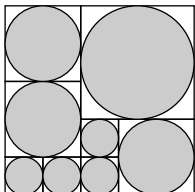
SOLUTION: Equivalently we need the integers in $(-\sqrt{21}, \sqrt{21})$. As $4 < \sqrt{21} < 5$, clearly these integers in increasing order are the numbers $-4, -3, \dots, 3, 4$. There are 9 of them.

3. A cube with edge 1 is cut into two identical cuboids. What is the surface area of one of these cuboids?

- (A) $\frac{3}{2}$ (B) 2 (C) 3 (D) 4 (E) 5

SOLUTION:

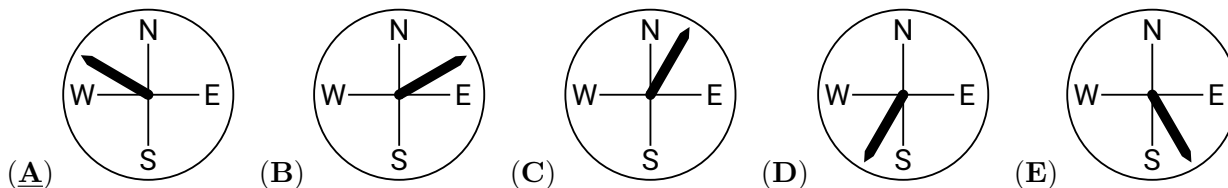
4. A large square is divided into smaller squares, as shown. A shaded circle is inscribed inside each of the smaller squares. What proportion of the area of the large square is shaded?



- (A) $\frac{8\pi}{9}$ (B) $\frac{13\pi}{16}$ (C) $\frac{3}{\pi}$ (D) $\frac{3}{4}$ (E) $\frac{\pi}{4}$

SOLUTION: Observe that a circle inside a square and the square itself have a constant proportion of areas, independent of the length of the side. This constant is easily seen to be $\frac{\pi}{4}$. So the same constant is the answer to the problem.

5. After the storm last night, the flagpole on our school building is leaning over. Looking from northwest, its tip is to the right of its bottom point. Looking from the east, its tip is also to the right of its bottom point. In which direction could the flagpole be leaning over?



SOLUTION: Consider the ground on which the flagpole stands as a co-ordinate plane with the base of the flagpole at the origin, the x-axis running from West to East and the y-axis from South to North. Looking from northwest we see that the top of the pole is below the line $y = -x$ while looking from the east we see that it is above the x-axis. Only answer (A) satisfies both of these conditions.

6. A rectangular sheet of paper has length x and width y , where $x > y$. The rectangle may be folded to form the curved surface of a circular cylinder in two different ways. What is the ratio of the volume of the longer cylinder to the volume of the shorter cylinder?

- (A) $y^2 : x^2$ (B) $y : x$ (C) 1:1 (D) $x : y$ (E) $x^2 : y^2$

SOLUTION:

7. Let $x = \frac{\pi}{4}$. Which of the following numbers is the largest?

- (A) x^4 (B) x^2 (C) x (D) $\sqrt{x}$ (E) $\sqrt[4]{x}$

SOLUTION:

8. How many 3-digit-numbers formed using only the digits 1, 3 and 5 are divisible by 3? You may use digits more than once.

- (A) 3 (B) 6 (C) 9 (D) 18 (E) 27

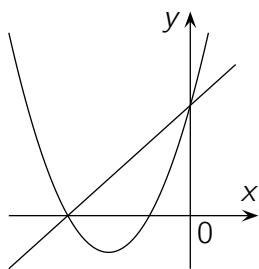
SOLUTION:

9. What is the area of the triangle whose vertices are at (p, q) , $(3p, q)$ and $(2p, 3q)$, where $p, q > 0$?

- (A) $\frac{pq}{2}$ (B) pq (C) $2pq$ (D) $3pq$ (E) $4pq$

SOLUTION:

10. The parabola in the figure has an equation of the form $y = ax^2 + bx + c$ for some distinct real numbers a , b and c . Which of the following equations could be an equation of the line in the figure?



- (A) $y = bx + c$ (B) $y = cx + b$ (C) $y = ax + b$ (D) $y = ax + c$ (E) $y = cx + a$

SOLUTION: From the shape of the graph we realise that the parabola has two distinct negative roots and $a > 0$. It follows that also $b, c > 0$. Since a and b are different from c , we can discard (B), (C) and (E). Let's consider (A): if $y = bx + c$ were the correct answer, then $x = -c/b$ would be a root of the quadratic. Test this root on the original equation and it will lead to $ac^2 = 0$, which is a contradiction as $a, c > 0$. Now we have just to see that (D) is possible. An example: $ax^2 + (a + c)x + c$ with roots -1 and $-c/a$ where $c > a > 0$ satisfies all requirements.

4 points

11. What proportion of all the divisors of $7!$ is odd?

- (A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{1}{4}$ (D) $\frac{1}{5}$ (E) $\frac{1}{6}$

SOLUTION: Solution 1) The number is of the form 2^4a , with a odd. For every odd divisor b of $7!$ (or, equivalently, of a) we have four even ones, namely $2b$, 2^2b , 2^3b and 2^4b . So odd divisors are $1/5$ of all. Solution 2) The number is $2^4 \cdot 3^2 \cdot 5 \cdot 7$. Note odd divisors are the divisors of $3^2 \cdot 5 \cdot 7$. From the formula for the number of divisors we see that the total number of divisors $(4 + 1)(2 + 1)(1 + 1)(1 + 1) = 60$ while the number of odd divisors is $(2 + 1)(1 + 1)(1 + 1) = 12$. Hence $1/5$ of the divisors are odd.

12. If $A = (0, 1) \cup (2, 3)$ and $B = (1, 2) \cup (3, 4)$, what is the set of all numbers of the form $a + b$ with a in A and b in B ?

- (A) $(1, 7)$ (B) $(1, 5) \cup (5, 7)$ (C) $(1, 3) \cup (3, 7)$
 (D) $(1, 3) \cup (3, 5) \cup (5, 7)$ (E) none of the previous

SOLUTION:

13. How many three-digit natural numbers have the property that when their digits are written in reverse order, the result is a three-digit number which is 99 more than the original number?

- (A) 8 (B) 64 (C) 72 (D) 80 (E) 81

SOLUTION: It is given that $100C + 10B + A - (100A + 10B + C) = 99$ so $99(C - A) = 99$. It follows that $C = A + 1$. So the numbers have the form $1B2$ or $2B3$ or $3B4$ or ... or $8B9$ (eight cases). Since B could be any of $0, 1, 2, \dots, 9$ (ten cases), there is a total of $8 \cdot 10 = 80$ numbers with the given property.

14. The first 1000 positive integers are written in a row in some order and all sums of any three adjacent numbers are calculated. What is the greatest number of odd sums that can be obtained?

- (A) 997 (B) 996 (C) 995 (D) 994 (E) 993

SOLUTION: Note that the total number of sums of any three adjacent numbers is 998. Firstly, we show that it is impossible for all sums to be odd. Assume it is possible. Then, the list of numbers must take one of the following forms:

Case 1. odd, odd, odd, ...

This is not possible since only odd numbers are used here.

Case 2. odd, even, even, odd, even, even, odd, even, even, ... This is not possible since we need more even numbers than odd numbers for this order.

Case 3. even, odd, even, even, odd, even, even, odd, even, ... As above this is not possible since we need more even numbers than odd numbers for this order.

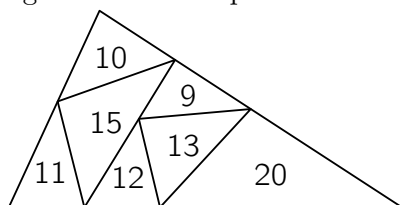
Case 4. even, even, odd, even, even, odd, even, even, odd, ... As above this is not possible since we need more even numbers than odd numbers for this order.

Thus, we come to a contradiction and at least one sum must be even. Now we show an example with one even sum only. For the following order all sums of the three consecutive numbers are odd with the only exception – the 750th sum below is even:

$$\underbrace{o, e, e, o, e, e, o, e, e, \dots, o, e, e}_{750 \text{ numbers}} \underbrace{o, o, o, \dots, o}_{250 \text{ numbers}}$$

Indeed, even+odd+odd=even. Thus, the greatest number of odd sums that can be obtained is 997.// A slightly different argument for the first part: First show that it is impossible to have all 998 sums odd. In this case the parity of the first three entries are periodically repeated (note the fourth entry is the same as the first, etc). For example e,o,e gives the entire sequence uniquely determined as (e,o,e)(e,o,e)(e,o,e)... Note now that as every periodically repeated part has three entries, at least two of them are same, a majority of 2 to 3 (at least). So for the entire sequence would have at least 2×333 numbers of the same parity, but we know they should be half. A contradiction.

15. A large triangle is divided into smaller triangles as shown. The number inside each small triangle indicates its perimeter. What is the perimeter of the large triangle?



(A) 31

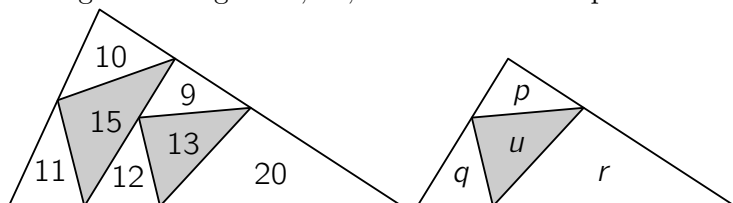
(B) 34

(C) 41

(D) 62

(E) none of the previous

SOLUTION: If we add the perimeters of the triangles that share common sides with the outer triangle (these are the triangles of perimeters 10, 9, 20, 12 and 11) we get the perimeter of the large triangle increased by the inner segments. But these segments add up to the perimeters of the inner triangles. So if we subtract them, the perimeter of the large triangle is $(10 + 9 + 20 + 12 + 11) - (15 + 13) = 34$. Another way to solve the problem is to observe (see figure on the right) that the perimeter of the triangle is $p + q + r - u$. Applying this twice to the original figure we find that the triangle containing the 9, 20, 12 and 13 triangles has perimeter $9 + 20 + 12 - 13 = 28$. Now view the original triangle as containing the triangles 11, 28, 10 and 15. So its perimeter is $11 + 28 + 10 - 15 = 34$.



16. For a positive integer N , we denote by $p(N)$ the product of the digits of N when written in decimal form. For example, $p(23) = 2 \times 3 = 6$. What is the value of the sum $p(10) + p(11) + p(12) + \dots + p(99) + p(100)$?

- (A) 2025 (B) 4500 (C) 5005 (D) 5050
(E) none of the previous

SOLUTION: Note that numbers that contain a 0 do not contribute to the sum. For the numbers from 10 to 19 the contribution to the sum is $1 \times 0 + 1 \times 1 + \dots + 1 \times 9 = 1 \times (1 + 2 + \dots + 9) = 45$. Similarly the contribution of 20 to 29 is 2×45 , and so on up to 90 to 99 that contribute 9×45 . The total is $1 \times 45 + 2 \times 45 + \dots + 9 \times 45 = (1 + 2 + \dots + 9) \times 45 = 45 \times 45 = 2025$

17. In the 5×5 square shown the sum of the numbers in each row and in each column is the same. There is a number in every cell, but some of the numbers are not shown. What is the number in the cell marked with a question mark?

	16		22	
20		21		2
	25		1	
24		5		6
	4		?	

- (A) 8 (B) 10 (C) 12 (D) 18 (E) 23

SOLUTION: The sum of rows a and b is the same as the sum of the columns c and d . Note that these two sets share four numbers (they are the ones marked X in the figure shown), so we can ignore them. What remains must also have the same sum, so from $a + b = c + d$ we conclude $(20 + 21 + 2) + (24 + 5 + 6) = (16 + 25 + 4) + (22 + 1 + x)$. So $x = 10$. If we want to see a magic square completed, the one on the right would do. It also has the same sum, namely 65, on the two diagonals as well (although for the problem this was not a requirement).

			c		d						
			16		22		3	16	9	22	15
b	20	$\times$		21	$\times$	2	20	8	21	14	2
			25		1		7	25	13	1	19
a	24	$\times$		5	$\times$	6	24	12	5	18	6
			4		?		11	4	17	10	23

18. A piece of string is lying on the table. It is partially covered by three coins as seen in the

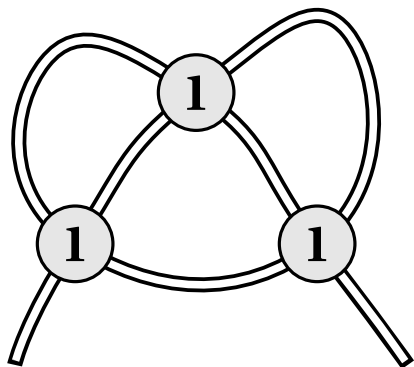


figure.



Under each coin the string is equally likely to pass over itself like this:



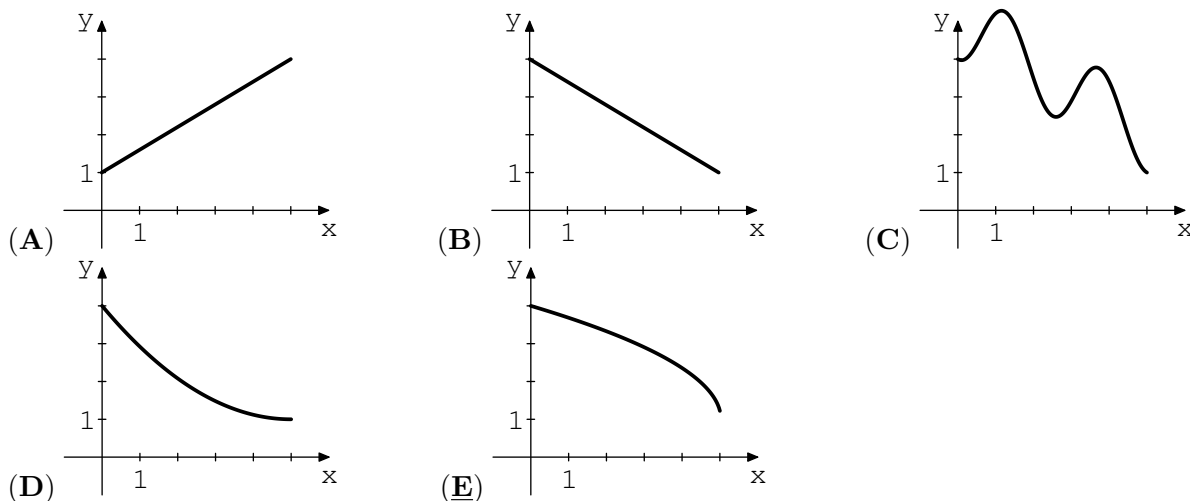
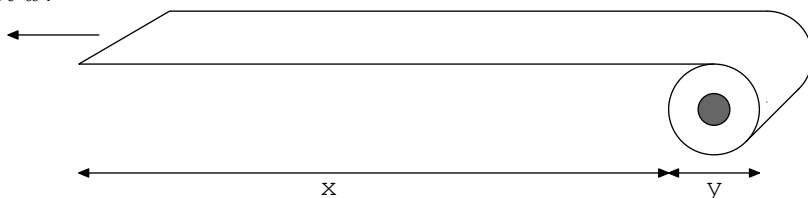
or like this:

What is the probability that the string is knotted after its ends are pulled?

- (A) $\frac{1}{2}$ (B) $\frac{1}{4}$ (C) $\frac{1}{8}$ (D) $\frac{3}{4}$ (E) $\frac{3}{8}$

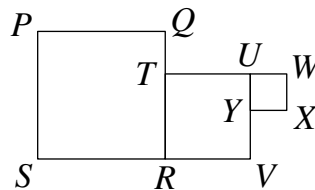
SOLUTION: Without loss of generality a certain position can be assumed under the top coin. That leaves 4 cases to check, and only one of them contains a knot. Another way: following the string from one end, the crossings must be above-below-above-below-above-below or below-above-below-above-below-above, so $2/8 = 1/4$.

19. A naughty pup grabs the end of a roll of toilet paper and walks away at a constant speed. Which of the functions below best describes the thickness y of the roll as a function of the unrolled part x ?



SOLUTION:

20. The diagram shows three squares, $PQRS$, $TRVU$ and $UWXY$. They are placed together, edge to edge. Points P , T and X lie on the same straight line. The area of $PQRS$ is 36 and the



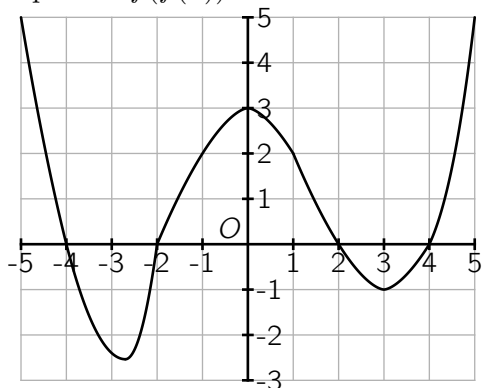
area of $TRVU$ is 16. What is the area of triangle PXV ?

- (A) $14\frac{2}{3}$ (B) $15\frac{1}{3}$ (C) 16 (D) $17\frac{2}{3}$ (E) 18

SOLUTION:

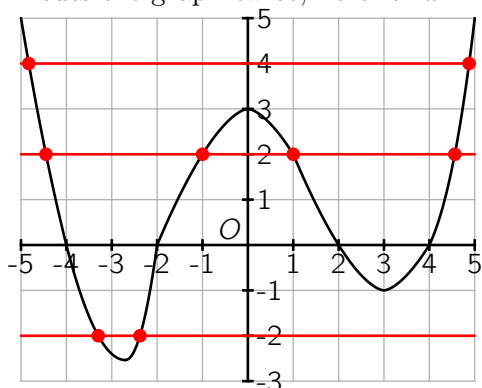
5 points

21. The figure shows the graph of a function $f : [-5, 5] \rightarrow \mathbb{R}$. How many distinct solutions does the equation $f(f(x)) = 0$ have?



- (A) 2 (B) 4 (C) 6 (D) 7 (E) 8

SOLUTION: From the graph we conclude that the roots of f are -4 , -2 , 2 and 4 . So if $f(f(x)) = 0$, which means that $f(x)$ is a root of f , we have $f(x) = -4$ or $f(x) = -2$ or $f(x) = 2$ or $f(x) = 4$. We solve these separately: The equation $f(x) = -4$ has no roots because the line $y = -4$ does not cut the graph. The equation $f(x) = -2$ has two roots because the line $y = -2$ cuts the graph twice. Namely for $x = -3$ and for $x \approx -2.5$ (this last value is only rough guides as read from the graph). The equation $f(x) = 2$ has four roots because the line $y = 2$ cuts the graph four times, namely for $x \approx -4.5$, $x = -1$, $x = 1$ and $x \approx 4.5$. Finally, the equation $f(x) = 4$ has two roots because the line $y = 4$ cuts the graph twice, here for $x \approx -4.8$ and $x \approx 4.8$. Summing up, we have $2+4+2=8$ roots.



22. The numbers 1, 2, 7, 9, 10, 15 and 19 are written down on a blackboard. Two players alternately delete one number each until only one number remains on the blackboard. The sum of the numbers deleted by one of the players is twice the sum of the numbers deleted by the other player. What is the number that remains?

- (A) 7 (B) 9 (C) 10 (D) 15 (E) 19

SOLUTION: The sum of all deleted numbers by both players is divisible by 3. The sum of all given numbers is 63 and is divisible by 3 too. It follows that the remaining number is 15 or 9. But the case it is 15 does not work and the only possibility is that the remaining number is 9. There is a unique realization: 1, 2 and 15 are the deleted numbers by the one player while 7, 10 and 19 are the deleted numbers by the second player.

23. The function $f(x)$ is such that $f(x+y) = f(x) \cdot f(y)$ and $f(1) = 2$. What is the value of $\frac{f(2)}{f(1)} + \frac{f(3)}{f(2)} + \dots + \frac{f(2021)}{f(2020)}$?

- (A) 0 (B) $\frac{1}{2}$ (C) 2 (D) 2020
(E) none of the previous

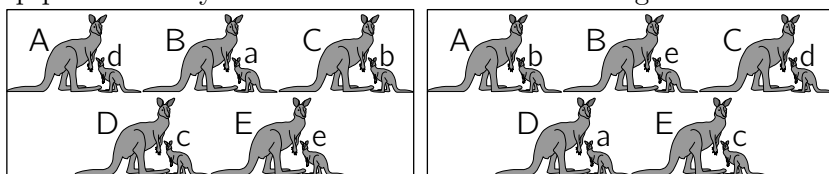
SOLUTION: For any x, y , we have $f(x+y) = f(x) \cdot f(y)$. So, let $y = 1$, then $f(x+1) = f(x) \cdot f(1)$, $\frac{f(x+1)}{f(x)} = f(1)$. Hence, let $x = 1, 2, 3, 4, 5, \dots, 2020$, respectively, we can get

$$\frac{f(2)}{f(1)} = \frac{f(3)}{f(2)} = \dots = \frac{f(2021)}{f(2020)} = f(1) = 2$$

Hence,

$$\frac{f(2)}{f(1)} + \frac{f(3)}{f(2)} + \dots + \frac{f(2021)}{f(2020)} = 2 \times 2020 = 4040.$$

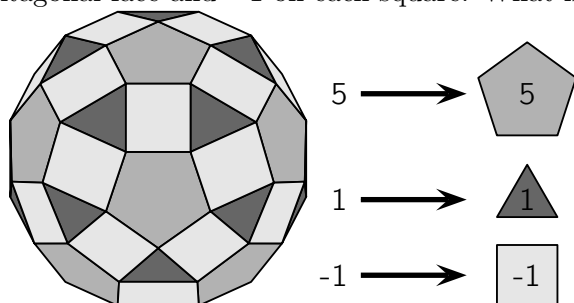
24. Five kangaroos named A, B, C, D and E have one child each, named a, b, c, d and e. In the first group photo shown exactly two of the children are standing next to their mothers. In the second group photo exactly three of the children are standing next to their mothers. Whose child is a?



- (A) A (B) B (C) C (D) D (E) E

SOLUTION: The mother/child pairs are $A/d, B/e, C/b, D/a, E/c$. HINT: In the two photos there are two cycles of children. One is $d \rightarrow b$ and back, and the other is $a \rightarrow e \rightarrow c$ and back.

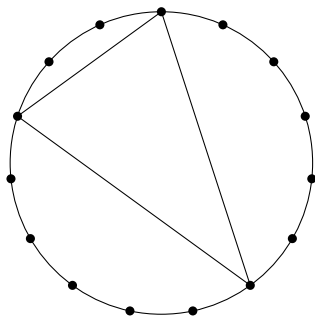
25. The solid shown in the diagram has 12 regular pentagonal faces, the other faces being either equilateral triangles or squares. Each pentagonal face is surrounded by 5 square faces and each triangular face is surrounded by 3 square faces. John writes 1 on each triangular face, 5 on each pentagonal face and -1 on each square. What is the total of the numbers written on the solid?



- (A) 20 (B) 50 (C) 60 (D) 80 (E) 120

SOLUTION: Each square shares one side with two distinct pentagons and two distinct triangles. So there are $12 \times 5 \div 2 = 30$ squares and $30 \times 2 \div 3 = 20$ triangles. $12 \times 5 + 30 \times (-1) + 20 \times 1 = 50$.

26. On a circle 15 points are equally spaced. We can form triangles by joining any three of these. We count two triangles as being the same if they are congruent i.e. one is a rotation and/or a reflection of the other. How many different triangles can be drawn?



- (A) 19 (B) 91 (C) 46 (D) 455 (E) 23

SOLUTION: There are $\binom{15}{3} = 455$ ways of choosing three points.

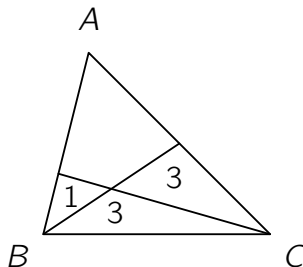
There are 5 congruent equilateral triangles in these 455, which we count as 1.

Of the remaining 450, there are 6 different isosceles triangles, each in 15 different rotations. So there are 90 isosceles triangles, which we count as 6.

The remaining 360 triangles are scalene, so occur in 15 different rotations and 15 different reflections. So these only count for $360 \div 30 = 12$ triangles.

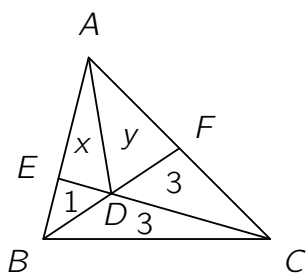
In all there are $1 + 6 + 12 = 19$ different triangles.// Another solution: The circle is divided into 15 arcs of $360:15=24$ degrees each. As each of the required triangles has its angles standing on one or more such arcs, its angles are either $24:2=12$ degrees or multiples of 12 degrees. So for each triangle there are positive integers $p \leq q \leq r$ such that $12p+12q+12r=180$, so $p+q+r=15$. In other words we need to count the integer partitions of 15 into three parts. Note that each partition will give rise to a different triangle because their angles are different (so we do not have to worry about rotations or reflections). But such partitions of 15 are easy to write if we do it systematically starting from $p=1$ and varying q and r , then $p=2$ and varying q and r , etc. We will find that there are 19 of them, namely $1+1+13, 1+2+12, 1+3+11, \dots, 1+7+7, 2+2+11, 2+3+10, \dots, 2+6+7, 3+3+9, \dots, 3+6+6, 4+4+7, 4+5+6, 5+5+5$.

27. A triangle ABC is divided into four parts by two straight lines, as shown. The areas of the smaller triangles are 1, 3 and 3. What is the area of the original triangle?

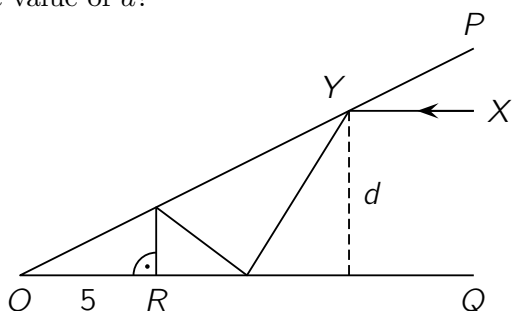


- (A) 12 (B) 12.5 (C) 13 (D) 13.5 (E) 14

SOLUTION: Let the areas of the two triangles shown be x and y . From the two triangles of area 3 we conclude that $BD = DF$. So $(ABD) = (ADF)$, hence $x + 1 = y$. Similarly $(BDC) = 3(BDE)$, so $DC = 3ED$ and so $y + 3 = 3x$. Solving the system we find $x = 2, y = 3$ which means the original triangle has area $1 + 3 + 3 + 3 + 2 = 12$.



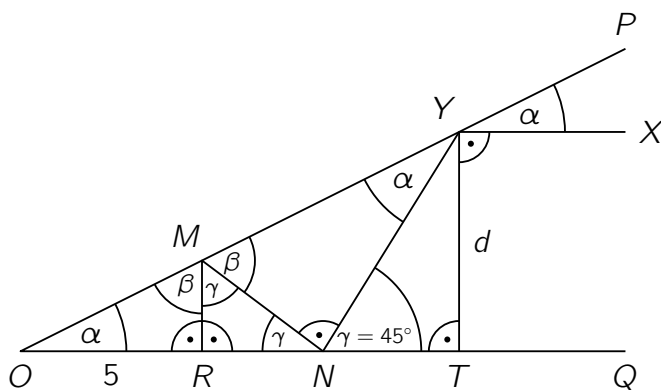
28. Two plane mirrors OP and OQ are inclined at an acute angle (diagram is not to scale). A ray of light XY parallel to OQ strikes mirror OP at Y . The ray is reflected and hits mirror OQ , is reflected again and hits mirror OP and is reflected for a third time and strikes mirror OQ at right angles at R , as shown. The distance OR is 5 cm. The ray XY is d cm from the mirror OQ . What is the value of d ?



- (A) 4 (B) 4.5 (C) 5 (D) 5.5 (E) 6

SOLUTION: Let the mirrors be inclined at an angle α . By assumption the ray strikes the first time the mirror OP at angle α . Let β and γ be the angles at which the ray strikes again the mirrors, as in the picture. The triangles ORM and YNM are similar, hence $\gamma = 45^\circ$. From this we get $|MN| = \sqrt{2}|MR|$ and $|NY| = \sqrt{2}|TY|$. Taking into account again similarity of the triangles ORM and YNM we have

$$\frac{|MR|}{5} = \frac{|MN|}{|NY|} = \frac{\sqrt{2}|MR|}{\sqrt{2}|TY|} \text{ i.e. } |TY| = d = 5.$$



Alternative solution - We have three angles α , two β and two γ as shown. It follows that the triangle YNM is a right angled triangle because so is, by assumption, ORM . So looking in turn, from right to left, at the four right angled triangles we have

$$d = |TY| = |NY| \sin \gamma = \frac{|MN| \sin \gamma}{\tan \alpha} = \frac{|MR|}{\tan \alpha} = |OR| = 5,$$

and we are done.

29. Let $M(k)$ be the maximum value of $|4x^2 - 4x + k|$ for x in the interval $[-1, 1]$, where k can be any real number. What is the minimum possible value of $M(k)$?

- (A) 4 (B) $9/2$ (C) 5 (D) $11/2$ (E) 8

SOLUTION: The vertex of the parabola represented by the equation $f(x) = 4x^2 - 4x + k$ has $x = \frac{1}{2}$. The maximum absolute value of this parabola will occur either at one end of the interval or at the vertex. Consequently, for each $k \in \mathbb{R}$, we have $M(k) = \max\{|8 + k|, |k - 1|, |k|\}$, and the minimum value of $M(k)$ is $\frac{9}{2}$, achieved for $k = -\frac{7}{2}$.

30. A certain game is won when one player gets 3 points ahead. Two players A and B are playing the game and at a particular point, A is 1 point ahead. Each player has an equal probability of winning each point. What is the probability that A wins the game?

- (A) $\frac{1}{2}$ (B) $\frac{2}{3}$ (C) $\frac{3}{4}$ (D) $\frac{4}{5}$ (E) $\frac{5}{6}$

SOLUTION: After the first point played, the score is either +2 to A or even. If it is even, each player has a 50% chance of winning from that point on. If it is +2 to A, then on the next point, either A wins or we are back where we started with equal probability. Thus we have a $\frac{1}{4}$ chance of A winning in the next two points, a $\frac{1}{4}$ chance of A winning despite losing the next point and a $\frac{1}{4}$ chance of B winning after winning the next point. The remaining $\frac{1}{4}$ is the probability of being back where we started after 2 points, so the ratio of A winning to B winning is 2:1. Hence the probability that A wins is $\frac{2}{3}$.

Alternative solution: Let p be the probability of A winning. We look at the next two games. They can be AA, AB, BA and BB, each with probability $1/4$. In the first case A wins. In the second and in the third case we are in exactly in the same situation as at the beginning (p). In the fourth we are as at the beginning but with the roles of A and B reversed. This means B has chance p of winning so A has $1 - p$. Summing we get:

$$p = 1/4 + p/4 + p/4 + (1 - p)/4, \text{ so } p = 2/3.$$